



cacheAdapter/Hub/RepeaterModule User Manual

- V1.0



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1. Product Overview

1.1 Product Introduction

MR0-485-2M1 module is a two-way 485 serial port to one-way 485 serial port communication module, which can realize the function of two master stations accessing one slave station, and realize the communication between the master station and the slave station.

It has the function of communicating at different baud rates between stations. It is an economical, stable, easy to install and highly applicable product.

The MR0-485-1T2 module is a one-way 485 serial port to two-way 485 serial port communication module, which realizes the isolation function of one to two and extends the communication distance.

2. Product Specifications

2.1、Interface parameters of each product

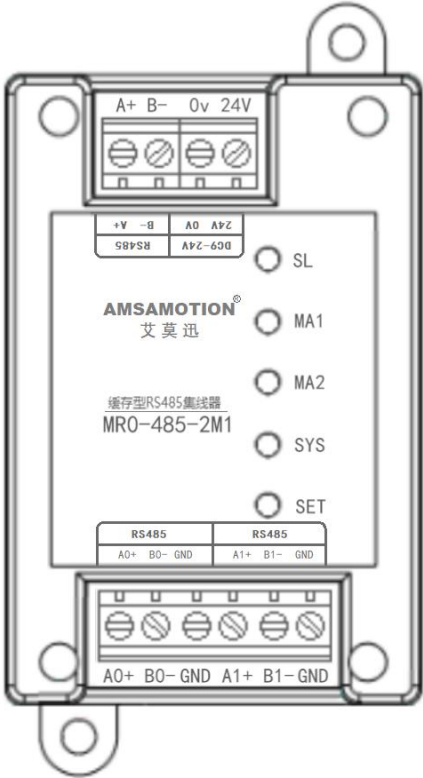
MR0-485-2M1 Main Parameters	
Slave serial port parameters (RS485 communication parameters)	
Interface Type	RS485 (for wiring terminals)
Baud rate	1200~256000 (default 9600, configurable)
Communication Format	Default 8-bit data, 1-bit stop, no parity (configurable)
Transmission distance	When the baud rate is 9600, the 485 serial communication distance is 1200 meters, whichever is greater.
Master station serial port 1 parameters (RS485 communication parameters)	
Interface Type	RS485 (for wiring terminals)
Baud rate	1200~256000 (default 9600, configurable)
Communication Format	Default 8-bit data, 1-bit stop, no parity (configurable)
Transmission distance	When the baud rate is 9600, the 485 serial communication distance is 1200 meters, whichever is greater.
Master station serial port 2 parameters (RS485 communication parameters)	
Interface Type	RS485 (for wiring terminals)
Baud rate	1200~256000 (default 9600, configurable)
Communication Format	Default 8-bit data, 1-bit stop, no parity (configurable)
Transmission distance	When the baud rate is 9600, the 485 serial communication distance is 1200 meters, whichever is greater.
Power supply parameters	
Operating voltage	DC 24V; with anti-reverse polarity protection
Power consumption	Less than 0.2W

Work Environment	
Operating temperature	-10°C~+50°C
Storage temperature	-20°C~+70°C
other	
Installation	guide
Dimensions	82*54*32mm

MR0-485-1T2 Main Parameters	
Master station serial port parameters (RS485 communication parameters)	
Interface Type	RS485 (for wiring terminals)
Baud rate	1200~256000 (adaptive)
Slave serial port 1 parameters (RS485 communication parameters)	
Interface Type	RS485 (for wiring terminals)
Baud rate	1200~256000 (adaptive)
Slave serial port 2 parameters (RS485 communication parameters)	
Interface Type	RS485 (for wiring terminals)
Baud rate	1200~256000 (adaptive)
Power supply parameters	
Operating voltage	DC 24V; with anti-reverse polarity protection
Power consumption	Less than 0.2W
Work Environment	
Operating temperature	-10°C~+50°C
Storage temperature	-20°C~+70°C
other	
Installation	guide
Dimensions	82*54*32mm

2.2 Interface Description

2.2.1 Terminal Description



Terminal number	Functional Description	Terminal number	Functional Description
24V	DC 24V power supply positive terminal	B0-	Master 1 Serial port 485 B
0V	Function 1: DC 24V power negative pole	GND	Serial port ground terminal
	Function 2: 485 communication ground terminal	A1+	Master 2 Serial Port 485 A
A+	Slave serial port 485 A	B1-	Master 2 Serial Port 485 B
B-	Slave serial port 485 B	GND	Serial port ground terminal
A0+	Master 1 Serial port 485 A		

2.2.2、Indicator Light Description

name	illustrate
SL	Slave serial port receiving and sending signal light
MA1	Master station 1 serial port transceiver indicator
MA2	Master station 2 serial port transceiver signal light

SYS	System indicator light: flashes slowly for one second when the module is normal. Flashes quickly when entering parameter setting mode, flashes quickly for 5 seconds when operating Reset, and then turns to normal flashing (only MR0-485-2M1 module has
SET	Set serial port parameters and reset button (only available for MR0-485-2M1 module)

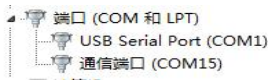
3. MR0-485-2M1 User Manual

3.1. Use the “Analog Configuration Tool”

3.1.1. Preparation before using the configuration tool

- Use USB to 485 serial cable to connectModule slave serial portand computer, confirm the COM port (from "My Computer - Properties / Management - Device Management")

Confirm the COM port by clicking on "Device-Port"



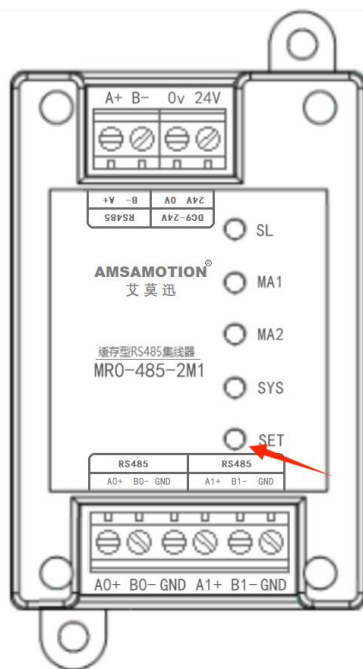
- Connect the DC 24V external power supply to the module and power on. Before powering on, please check whether the positive and negative poles of the power supply are connected correctly.

- Open "RAEC_RS485_Tool"

3.1.2. Configuration tool connection steps

To use the configuration tool, you need to establish communication between the host computer and the module. Please follow the steps below to connect the configuration tool:

A. Within one minute after power on, press SET and the SYS light will flash quickly, indicating that you have entered the configuration parameter mode and can then configure the parameters (press to exit the configuration parameter mode, and the SYS light will resume flashing normally)



B. Select the serial port number, that is, the port number of the USB to 485 serial cable connected to the module in the computer device manager.

If the corresponding port number is not found in the list, click the "Search Serial Port" button and select again

C. Select the baud rate, parity bit, data bit, and stop bit. They should be consistent with the current parameters of the module. No selection is required for factory or reset defaults.

D. Click the Open Serial Port button. After the connection is successful, the "Disconnect" button changes from purple to green, indicating that the communication connection is successful.

Form_MODBUS_2M1

选择连接参数

串口号:

波特率: 9600

校验位: None

数据位: 8

停止位: 1

打开串口

搜索串口

设备连接状态

固件版本:

读取参数

一键配置所有参数

参数配置

从站口参数配置

波特率:

数据位:

停止位:

校验位:

设置

主站口1通信参数配置

波特率:

数据位:

停止位:

校验位:

设置

主站口2通信参数配置

波特率:

数据位:

停止位:

校验位:

设置

主站口1超时时间配置

ms

设置

主站口2超时时间配置

ms

设置

2M1_V0.0.1

- COM port communication parameter description

COM port (485) communication parameter type			
Baud rate	Data bits	Stop bits	Check digit
1200~256000	8 bits (fixed)	1 (fixed)	None

- COM port communication parameter settings

After completing 3.1.2 communication connection, in the "Parameter Configuration" area of the configuration tool, select the "Baud Rate" and "Parity Bit" of the corresponding serial port.

In the list, select the required parameters and click "Write Configuration". The settings will take effect after power cycle.

Within 60 seconds after the module is powered on, press and hold the Reset button, release the button after 3 seconds (the SYS light will be on), then the SYS light will flash quickly.

After the SYS indicator resumes slow flashing, power off the module for at least 3 seconds and then power it on again. All serial ports will then be restored to factory settings, as shown in the following table.

Parameter name	Parameter default values
Baud rate	9600
The data is	8
Stop bits	1
Check digit	NONE
Master port 1 timeout	500ms
Master port 2 timeout	500ms

3.4、Serial port timeout description

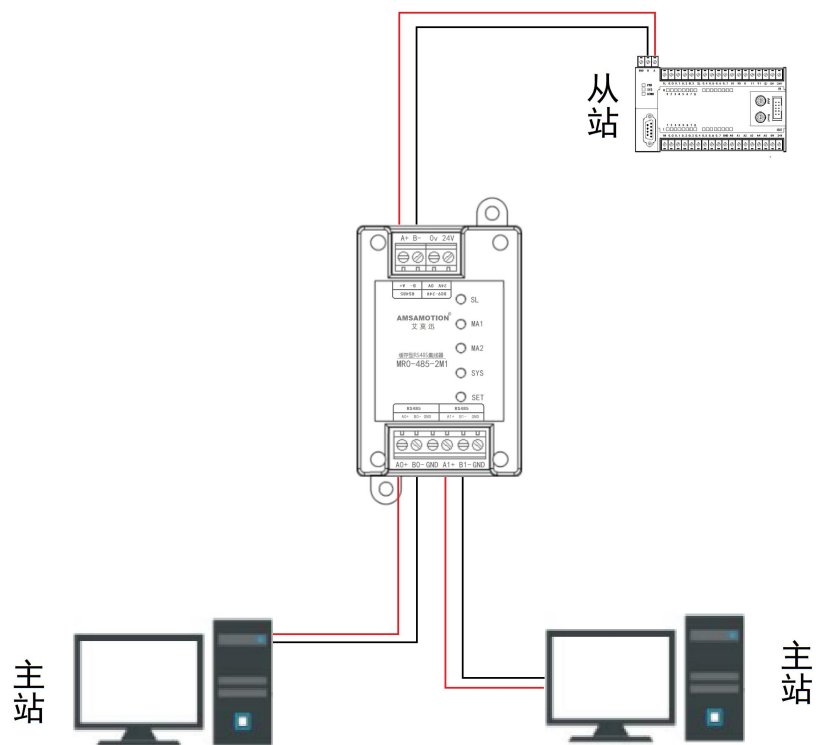
The module is a cache hub with a maximum cache of 20 instructions. The instructions of the two master stations enter the cache one after another, and the first instruction is sent from the serial port of the slave station.

After the timer is reached, the timer will start timing. If the timer exceeds the timeout period set by the master serial port that issued the command, the master serial port will not wait any longer and will continue to send the next command.

The station device responds and sends the next command immediately after answering the master station serial port that issued the command.

3.5 Wiring Diagram

The slave serial port can connect multiple slaves

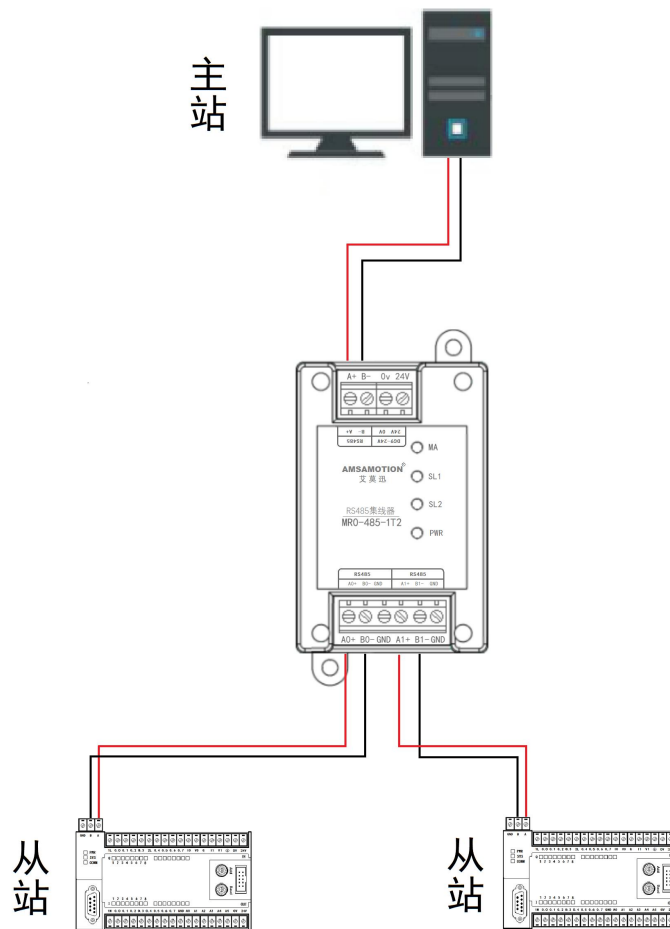


IV. MR0-485-1T2 User Manual

When using this module, please note that the serial port baud rate of the master station and each slave station must be consistent.

4.1 Wiring Diagram

The slave serial port can connect multiple slaves



Revision History

Version	Revision Date	Revision Notes	Maintainer
1.0	2024.12.26	Initial version	LZY